

Revised Generation Interconnection System Impact Study Report

For Queue Project Q14-0411

Commonwealth Chesapeake 138 kV
December 28, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0411. The Interconnection Customer proposes a 44.8 MW generating facility interconnecting at the Commonwealth Chesapeake 138 kV substation in the Virginia area. The request was received on December 10, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 248 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$599,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Easton 2 to Cady 345kV, the Shoals Substation to Bristol 138kV loads to 164.8% of its 164 MVA rating. Project DFAX: 7.2%.

For the loss of Bond-Elk Grove Florin 345-138 kV XFMR, the Edgewater Station to C W Tippy 138kV loads to 102.1% of its 301 MVA rating. Project DFAX: 7.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Shols to Bristol 138kV Line	Rebuild line with higher-capacity conductor	\$5,703,000	10% (\$581,000)
Edgewater Station to C W Tippy 138kV	Reconductor line	\$1,001,000	2% (\$18,000)
	Total allocated to project		\$599,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.